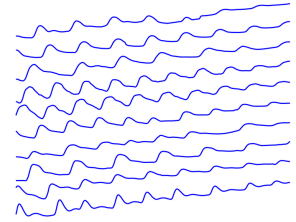


Photoplethysmography Datasets



https://peterhcharlton.github.io/post/ppg_datasets/

Here's a summary of the photoplethysmography datasets grouped by type:

Clinical & Critical Care

- **MIMIC Critical Care Database** (10,000s subjects): Recordings from critically-ill adults and neonates with ECG, BP, and respiratory signals
- **MIMIC PERform Training and Testing** (400 subjects): 10-minute recordings from critically-ill patients
- **MIMIC PERform AF Dataset** (35 subjects): Categorized by atrial fibrillation vs. normal sinus rhythm
- **BIDMC** (53 subjects): 8-minute recordings from critically-ill adults
- **CapnoBase** (42 subjects): Recordings during elective surgery and anaesthesia

Sleep & Polysomnography

- **MESA Dataset** (2,056 subjects): Finger PPG during polysomnography
- **SOMNIA Database** (100s subjects): Wrist PPG from children and adults during sleep studies
- **Sleep Disordered Breathing Database** (146 subjects): ~3-hour finger recordings from children

Healthy Subjects & Daily Activities

- **UK Biobank** (205,357 subjects): Single finger PPG from middle-aged subjects
- **PPG-DaLiA Data Set** (15 subjects): ~2.5 hours during daily living activities
- **WESAD Data Set** (15 subjects): ~2 hour protocol with amusement, stress, and relaxation
- **Real-World PPG dataset** (35 subjects): Multiple 6-second recordings from ages 10-74

Exercise & Physical Activity

- **VitalDB** (6,153 subjects): Finger PPG during operations
- **Vortal Dataset** (57 subjects): 10-min recordings before/after exercise
- **gyro-acc-ppg Dataset** (24 subjects): Wrist recordings during 12-minute exercise protocol
- **IEEE Signal Processing Cup 2015** (12 subjects): 5-min recordings during intensive exercise
- **Wrist PPG During Exercise** (8 subjects): Walking, running, and biking

Emotion & Stress

- **DEAP Database** (32 subjects): Thumb recordings while watching videos
- **ECSMP** (89 subjects): Wrist recordings designed to induce emotions
- **Eight-Emotion Sentics Dataset** (1 subject): 25-min emotion-evoking protocol over 20 days
- **AffectiveROAD Dataset** (10 subjects): ~86 minutes of driving recordings

Specialized Studies

- **PPG-BP Database** (219 subjects): Three finger recordings for blood pressure estimation
- **Pulse transit time during vasoconstriction Dataset** (86 subjects): Cold pressor and active stand tests
- **MARSH Dataset** (29 subjects): 15-min recordings during spontaneous and metronome-guided breathing
- **HYPE Dataset** (9 subjects): Wrist recordings from hypertensive subjects

Synthetic & Animal Data

- **Pulse Wave DataBase** (4,374 synthetic): Simulated PPG pulse waves
- **Photoplethysmography in dogs and cats** (21 animals): Recordings from dogs and cats

Impact Analysis:

The **highest impact and feasibility** datasets depend on your goal — here's a quick guide:

✓ Highest Impact + Feasibility (for most researchers & developers):

1. UK Biobank (Field 4205)

- **Impact:** Massive scale (205k+ subjects) → ideal for population-level studies, AI training, or epidemiological research.
- **Feasibility:** Publicly available, well-documented, no complex setup needed.
- **Best for:** Large-scale PPG analysis, cardiovascular risk prediction, or benchmarking algorithms.

2. MIMIC Critical Care Database

- **Impact:** Real-world ICU data with multimodal signals (ECG, BP, resp) → gold standard for clinical PPG applications.
- **Feasibility:** Requires registration (free for researchers), but widely used and well-supported.
- **Best for:** Critical care monitoring, sepsis detection, or PPG-based vital sign estimation.

3. PPG-DaLiA / WESAD

- **Impact:** Daily living + emotion/stress context → great for wearable AI, affective computing, or activity recognition.
- **Feasibility:** Small to medium size, clean annotations, easy to work with.
- **Best for:** Emotion detection, stress monitoring, or activity classification using wrist PPG.

🎯 Quick Recommendation:

If you want **maximum impact with minimal friction**, start with **UK Biobank**.

If you're targeting **clinical applications**, go with **MIMIC**.

For **wearable/emotion research**, **PPG-DaLiA or WESAD** are ideal.

Dataset	No. subs	Other signals	Description
UK Biobank (Field 4205)	205,357	None simultaneous. BP in same visit	Single finger PPG waves from middle-aged subjects.
MIMIC Critical Care Database	10,000s (growing)	ECG, BP, resp, others	Recordings from critically-ill adults and neonates, lasting from minutes to days. Typically at finger.
VitalDB	6,153	ECG, BP, resp, others	Finger PPG recordings from patients during operations.
MESA Dataset	2,056	ECG, resp, others	Finger PPG recordings from adults undergoing polysomnography.
MIMIC PERform Training and Testing	400	ECG, resp	Recordings from critically-ill adults and neonates, lasting 10 minutes. Typically at finger.
SOMNIA Database	100s (growing)	ECG, resp, others	Wrist PPG recordings from children and adults undergoing polysomnography.
PPG-BP Database	219	-	Three finger recordings from adults aged 20-89 with and without CVD, 3 waves per recording.
Sleep Disordered Breathing Database	146	-	Finger recordings lasting 3 hours, acquired from children referred for polysomnography.
ECSMP	89	ECG, accel, others	Wrist recordings acquired from healthy subjects during a protocol designed to induce different emotions.
Pulse transit time during vasoconstriction Dataset	86	ECG, BP	35-min wrist and finger recordings during cold pressor and active stand tests.
Vortal Dataset	57	ECG, resp	10-min finger and ear recordings before and after exercise from healthy adults aged 18-39 and >70.

<u>BIDMC</u>	53	ECG, BP, resp	8-min recordings from critically-ill adults (a subset of the MIMIC-II dataset).
<u>CapnoBase</u>	42	ECG, resp	8-min recordings from paediatrics and adults during elective surgery and anaesthesia.
<u>Bed-based BCG Dataset</u>	40	ECG, BCG, BP	Recordings from adults whilst at rest.
<u>MIMIC PERform AF Dataset</u>	35	ECG, resp	Recordings from critically-ill adults categorised as either AF (19 subjects) or normal sinus rhythm (16 subjects), lasting 10 minutes. Typically at finger.
<u>Real-World PPG dataset</u>	35	-	Recordings from healthy subjects aged 10 to 74 years old: several 6-second recordings per subject.
<u>DEAP Database</u>	32	ECG, resp, video, others	Thumb recordings from young, healthy subjects whilst watching one-minute 40 videos.
<u>University of Queensland Vital Signs Dataset</u>	32	ECG, resp, BP, EEG	Recordings from patients during anaesthesia, ranging from minutes to hours in duration.
<u>ESUM SNF Project Dataset</u>	31	accel, EDA	Wrist recordings whilst walking.
<u>MARSH Dataset</u>	29	ECG, resp	15-min finger recordings during spontaneous and metronome-guided breathing.
<u>Non-invasive BP Estimation</u>	26	ECG, BP, PCG	Finger recordings from healthy adults.
<u>gyro-acc-ppg Dataset</u>	24	ECG, accel, gyro	Wrist recordings from healthy subjects during an exercise protocol lasting 12 minutes.
<u>Pulse Transit Time PPG Dataset</u>	22	ECG, multiwav elength PPG	Finger recordings from healthy subjects during sitting, walking and running.
<u>Welltory-PPG-dataset</u>	21	RR intervals	Smartphone recordings acquired from the index finger in contact with the camera.

Wearable and Clinical Devices Dataset	18	ECG, resp, accel, EDA	Wrist PPG recordings acquired for 5 mins at rest, and 5mins whilst walking on spot.
PPG-DaLiA Data Set	15	ECG, resp, accel, EDA	Recordings acquired for 2.5 hours during a protocol of daily living activities.
WESAD Data Set	15	ECG, resp, accel, others	Recordings acquired in a 2 hour protocol designed to amuse, stress, and relax.
iAMwell Dataset	15	ECG, resp	20-min recordings before, during and after running.
Simultaneous Measurements Dataset	13	ECG, accel, resp	Recordings from adults at rest and during cognitive and physical tasks.
IEEE Signal Processing Cup 2015	12	ECG, accel, two PPGs	5-min recordings during intensive physical exercise from males aged 18-35.
AffectiveROAD Dataset	10	accel, EDA	Recordings acquired whilst driving a car for 86 minutes from mostly young adults.
Labeled raw PPG Signals	9	-	Finger recordings acquired at rest for 20-40 minutes from healthy adults.
HYPE Dataset	9	intermittent BP	Wrist recordings from hypertensive subjects: 8 subjects during stress tests, and 9 subjects during 24-hour monitoring.
Wrist PPG During Exercise	8	BP, accel	Wrist recordings acquired from adults aged 22-32 during walking, running and bike riding.
PPG-ACC Dataset	7	accel	Wrist recordings acquired from healthy adults aged 20-52 during rest, squatting and stepping.
Raw PPG Signal in Varying Levels of Activity	5	-	Finger recordings acquired at rest, talking and walking for 10 mins each from healthy adults.

PPG-Diary	1	two PPGs	A 28-day thumb recording from a healthy adult, with annotations of activities of daily living.
Eight-Emotion Sentic Dataset	1	EMG, GSR, resp	25-min recordings during a protocol to evoke emotions from 1 subject each day for 20 days.
Pulse Wave DataBase	4,374 (synthetic)	BP, blood flow, others	Single simulated PPG pulse waves representative of healthy adults aged 25-75.
Photoplethysmography in dogs and cats	21 (animals)	three PPGs	10-20 sec recordings at several arterial sites from 11 dogs and 10 cats.